L2.4 Limits at Infinity *Recall: L2.3 How Derivatives Affect the Shape of a Graph*

Name _____ **Solo – 9 minutes**

1. For $f(x) = x^3 - 6x^2 + 9x$, find the critical numbers and the sign of f' on each interval between them. Then classify each critical number by the First Derivative Test.

2. For $f(x) = x^3 - 6x^2 + 9x$, find every possible inflection point and the intervals of concavity. Then sketch f.

3. End behavior

(a) As x gets huge, what does $f(x) = x^3 - 6x^2 + 9x$ do?

(b) What does $g(x) = \dfrac{2x+1}{x-3}$ do as x goes to $+\infty$ or $-\infty$?